

The Two Diagnostics: Deficit Law at $k=3$ and $k=8,9$

$$d = m(n+1) - b \geq o \text{ — LRC in residue form}$$

An independent residue-proof architecture for four speeds (orbit lemma, divisor lemma, top-pair rescue), a counterexample hunt over 627,266 exhaustive speed sets at nine and ten speeds, and the bridge attribution that measures the residue reformulation against the factor-2 wall.

MIKEAA2020

lonely-runner project · computational-check
companion to the audit “the two diagnostics”

Zero violations at $k=8,9$ — 627,266 exhaustive sets, 155,382 adversarial evaluations, 143,892 bridge configs

Orbit lemma kills ~94–99% of the would-be-violation bridge at every size $n=3\dots10$ — stable, not decaying

Top-pair rescue law — coverage 92.8% → 98.6% (rising with n); the residual is eight parametric near-AP dilate families

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1 Introduction

For a finite set V of n distinct positive integer speeds, the loneliness of V is

$$\text{gap}(V) = \max_{t \in [0,1)} \min_{v \in V} \|vt\|,$$

where $\|x\|$ denotes the distance from x to the nearest integer. The Lonely Runner Conjecture (LRC) asserts $\text{gap}(V) \geq 1/(n+1)$ for every V . In the preceding audit (“the C-half attack and exhaustiveness”), the program reduced LRC, conditionally on a decomposition exhaustiveness statement, to the *deficit law*: at a reduced argmax $t^* = a/b$ of $f_V(t) = \min_v \|vt\|$, with $m = b \cdot \text{gap}(V)$, the integer

$$d = m(n+1) - b$$

satisfies $d \geq 0$. The deficit law is equivalent to $\text{gap}(V) \geq 1/(n+1)$; it is LRC in residue form. Its empirical status was: verified with zero violations on 20,804 + 297 crossings of the drop experiment, not proved; the covering argument of the previous turn reaches only $b \leq n(2m+1)+1$, a factor-2 short of the target $b \leq m(n+1)$.

The advisor’s assessment of that turn isolated the value of the program in the *frame* — the clean statement of the factor-2 obstruction — and prescribed two diagnostics, which this note executes completely.

D1. Independent residue proof at $k = 3$. Attempt to prove the deficit law for four speeds by residue methods, without importing the classical LRC proof. If the proof is genuinely shorter or more generalizable, the reformulation has content; if it merely re-derives the classical argument, the reformulation is a translation and the factor-2 obstruction is the only new thing.

D2. Counterexample hunt at $k = 8, 9$. Search for violations of the deficit law at nine and ten speeds, where LRC is known (Rosenfeld and Trakulthongchai) but not by residue methods. A failure would be a major finding (bug or counterexample); exceptions, if any, would be the interesting objects.

Between the two diagnostics we interpose a new experiment, the *bridge attribution* (Section 4), which measures — for every $n = 3, \dots, 10$ — how much of the would-be-violation region is closed by the b -grid residue arithmetic alone, and how much requires the off-grid crossing structure. This measurement is what turns the “clothing versus body” question into a number.

All computations use exact rational arithmetic. Every would-be violation flagged by the fast engine is re-verified with the repository’s independent reference solver (`lrc_gap_lib.gap_int`, the Lemma-6.1 candidate implementation) before being reported. Reproduction details are collected in Section 7.

2 The pair-sum candidate theorem

The engine behind both diagnostics rests on a sharpened form of the candidate theorem.

Theorem 2.1 (pair-sum candidates). *Let V be a set of $n \geq 2$ distinct positive speeds. The maximum of $f_V(t) = \min_{v \in V} \|vt\|$ over $[0, 1)$ is attained at some $t_0 = j/(u+w)$ with $u, w \in V$ and $0 \leq j < u+w$.*

Proof. The maximum is positive and attained in the open interval $(0, 1)$: $f(0) = 0$, and f is strictly positive at, e.g., an odd multiple of $1/(2v)$ minus an epsilon avoiding the finitely many zeros of the other speeds. Let $t_0 = a/b$ (in lowest terms) be a maximizing point and $A = \{v \in V : \|vt_0\| = f_V(t_0)\}$ the active set. In a neighborhood of t_0 the inactive speeds stay strictly above $f_V(t_0)$, so locally $f_V = \min_{v \in A} \|vt\|$.

Suppose first that some $v \in A$ sits at its pole, i.e. $vt_0 \equiv 1/2 \pmod{1}$. Then $f_V(t_0) = 1/2$ forces $\|v't_0\| \geq 1/2$, hence $\equiv 1/2$, for every $v' \in V$; thus $(v + v')t_0 \equiv 0 \pmod{1}$ for every pair, and t_0 lies on the $(v + v')$ -grid.

Otherwise all members of A are smooth at t_0 with signs determined by their residues $va \bmod b$: residue in $(0, b/2)$ gives slope $+v$, residue in $(b/2, b)$ gives slope $-v$. An interior local maximum cannot have all active slopes of one sign (all $+$ admits an increase to the right; all $-$ admits an increase to the left). Hence A contains u with residue $m_0 \in (0, b/2)$ and w with residue $b - m_0$, where $m_0 = b \cdot f_V(t_0)$. Then

$$(u + w)a \equiv m_0 + (b - m_0) \equiv 0 \pmod{b}.$$

Since $\gcd(a, b) = 1$, we get $b \mid u + w$, so $t_0 = a/b = j/(u + w)$ with $j = a(u + w)/b$. $\square$

Remark 2.2. The peak candidates $\text{odd}/(2v)$ and the difference-crossings $j/(w - u)$ of the Lemma-6.1 candidate list never carry a strict maximum alone: peaks force the all-poles case, which is on a pair-sum grid, and difference-crossings pair equal residues, i.e. equal slopes, which cannot bound a local maximum from both sides. The theorem was validated empirically: on 2,500 random sets the pair-sum-only engine agrees exactly with the reference implementation, and every reference argmax time satisfies the pair-sum property.

The engine (**GapEngine**) precomputes the distance tensor $T[j, D, v] = \min(vj \bmod D, D - vj \bmod D)$ over all pair sums $D \leq 2M + 1$ and reduces each evaluation to a gather and two reductions; it processes 10^5 – 10^6 exact gap evaluations per minute on two cores.

3 Diagnostic 2: the deficit law at nine and ten speeds

3.1 Exhaustive scans

Every n -subset of $[1..M]$ was scanned with the exact engine. Table 1 summarizes the outcome.

n	M	sets	min gap	violations	$d_{\min}$	critical sets (normalized)
3	18	816	1/4	0	0	1
4	18	3,060	1/5	0	0	2
5	18	8,568	1/6	0	0	2
6	18	18,564	1/7	0	0	1
7	18	31,824	1/8	0	0	3
8	18	43,758	1/9	0	0	1
9	20	167,960	1/10	0	0	1
10	21	352,716	1/11	0	0	1

Table 1: Exhaustive deficit-law scans: 627,266 sets, zero violations. “Critical sets” are gcd-normalized sets achieving $\text{gap} = 1/(n + 1)$ exactly. At $n = 4, 5, 7$ they are $\{1, \dots, n\}$ plus $\{1, 3, 4, 7\}$, $\{1, 3, 4, 5, 9\}$, $\{1, 2, 3, 4, 5, 7, 12\}$, $\{1, 4, 5, 6, 7, 11, 13\}$.

Two structural laws of the previous turn were re-verified at the argmax of every single set: the *pole law* (the residues $v_i a \bmod b$ attain both m and $b - m$) and the *covering wall* $b \leq n(2m + 1) + 1$: zero failures across all 627,266 sets, including all 520,676 sets at nine and ten speeds.

3.2 Adversarial search and critical isolation

A gap-minimizing random-restart hill-climb with plateau walks ($M \leq 60$) performed 102,122 evaluations at $n = 9$ and 53,260 at $n = 10$: the search bottoms out at exactly $1/10$ and $1/11$ respectively, with zero violations.

The critical census deserves its own statement. Non-regular critical sets (gap exactly $1/(n + 1)$, normalized set $\neq \{1, \dots, n\}$) exist at $n = 4, 5, 7$ inside $M = 18$; each is a floor-landing extension

of a rung-2 or tight core, the mechanism the drop experiment called a tight extension. A targeted hunt at $n = 8, 9, 10$ (growing all critical and rung-2 seeds at $n - 1$ and $n - 2$ by one or two fillers, $M \leq 60$) found *no* non-regular critical set; the repository’s tight-8 census at $V \leq 50$ independently contains exactly the six regular multiples $c\{1, \dots, 8\}$. The floor at $k = 8, 9$ is isolated to the regular family.

Remark 3.1 (isolation). The thinning of the critical census with n is the $k = 8, 9$ analogue of the “exception” question in D2: the deficit law holds with no exceptions, and the class of extremal configurations shrinks to a single point. Any proof of the deficit law at these sizes must, in residue terms, handle only the regular configuration and the shallow bridge above it (Section 4).

4 The bridge attribution: clothing versus body, measured

The *bridge region* for size n is

$$m(n+1) < b \leq n(2m+1) + 1 :$$

below it the deficit law holds trivially for a configuration; above it the covering union bound kills trivially. The bridge is where the factor-2 obstruction lives. A *bridge configuration* is a residue set $R = \{m\} \cup \{\text{interiors}\} \cup \{b-m\}$ at time a/b , with distinct interiors in $(m, b-m)$. Each configuration is tested at three levels:

- **L2 (orbit / b-grid):** some $j \in [1, b-1]$ has $\min_i \text{dist}(jr_i, 0) > m$ — a violation of the necessary condition of Lemma B below;
- **L3 (off-grid crossings):** for every unit $s \bmod b$, the canonical lift $c_i = sr_i \bmod b \in [1, b-1]$, viewed as a speed set, has gap $> m/b$;
- **realized:** some lift has gap $= m/b$ exactly — an actual violation of the deficit law.

Table 2 gives the kill profile across $n = 3, \dots, 10$ ($m \in \{1, 2, 3\}$ for $n \geq 5$, $m \in \{1, \dots, 8\}$ at $n = 4$; complete enumeration up to 500 configurations per (m, b) , seeded sampling beyond; all units for $n \leq 7$, 12 sampled units for $n \geq 8$; $b \leq 80$).

n	configs	L2 orbit kills	L3 off-grid kills	realized
3	192	186 (96.9%)	6	0
4	45,144	45,035 (99.8%)	109	0
5	14,075	13,781 (97.9%)	294	0
6	22,321	21,535 (96.5%)	786	0
7	27,923	26,611 (95.3%)	1,312	0
8	10,019	9,448 (94.3%)	571	0
9	11,414	10,725 (94.0%)	689	0
10	12,804	12,073 (94.3%)	731	0

Table 2: Bridge attribution: 143,892 would-be-violation configurations. The b -grid necessary condition (the residue “clothing”) kills a stable ~ 94 – 99% of the bridge at every size; the residue of the residual concentrates in the shallow bridge (small m , $|d| \leq 3$).

Three findings. First, *stability*: the orbit share does not decay with n ; the b -grid arithmetic keeps its covering power at every size. Second, *L2 anatomy*: about 55% of orbit kills are available at non-unit multipliers (the divisor lemma, Lemma C below), the rest need unit multipliers. Third, *L3 anatomy*: every killing crossing sits on a pair whose sum $D = u + w$ is not divisible by b (a pair with $D = b$ lies wholly on the b -grid and is capped by the orbit condition; $D = 2b$ is impossible for speeds below b); winners split between $D < b$ and $b < D < 2b$ roughly evenly, and winning values typically lie in $(m/b, 2m/b)$: the kills are decisive. The L3 survivors concentrate in the *shallow bridge* — small m and $|d| = b - m(n+1) \leq 3$ — which is to say: near-regular residue patterns. The deep bridge is entirely closed by the orbit condition.

5 Diagnostic 1: the residue proof architecture at $k = 3$

5.1 Two lemmas with proofs

Lemma 5.1 (orbit). *Let $t^* = a/b$ be an argmax of f_V with value m/b and residues $r_i = v_i a \bmod b$. Then for every $j \in \{1, \dots, b-1\}$,*

$$\min_i \text{dist}(j r_i, 0) \leq m.$$

Consequently $\bigcup_i \{j : \text{dist}(j r_i, 0) \leq m\} \supseteq \{1, \dots, b-1\}$ and therefore $b \leq n(2m+1) + 1$.

Proof. For every time t , $f_V(t) \leq \text{gap}(V) = m/b$. Apply this at $t = (ja/b) \bmod 1$: $\|v_i ja/b\| = \text{dist}(j r_i, 0)/b$, so $\min_i \text{dist}(j r_i, 0)/b \leq m/b$. The second statement is the union bound over the n sets $\{j : \text{dist}(j r_i, 0) \leq m\}$, each of size at most $2m+1$. $\square$

The union-bound corollary is the covering wall of the previous turn; Lemma 5.1 itself is strictly stronger, and Table 2 measures by exactly how much: the wall leaves the entire bridge open, while the pointwise orbit condition closes $\sim 94\%$ of it.

Lemma 5.2 (divisor). *Let d be a proper divisor of b with $dm < b$. Then d divides some residue r_i .*

Proof. Apply Lemma 5.1 at $j = b/d$. Writing $\rho_i = r_i \bmod d$, we have $(b/d)r_i \equiv (b/d)\rho_i \pmod{b}$, so $\text{dist}((b/d)r_i, 0) = (b/d) \cdot \min(\rho_i, d - \rho_i)$. For this to be at most m we need $\rho_i \leq dm/b$ or $\rho_i \geq d - dm/b$; under $dm/b < 1$ both force $\rho_i = 0$. $\square$

Example 5.3. At $n = 4$, $b = 12$, $m = 1$: the divisors 2, 3, 4, 6 of 12 must each divide one of the residues $\{1, r, s, 11\}$; since 11 is coprime to 12, one of r, s must be divisible by $\text{lcm}(4, 6) = 12$ — impossible inside $(1, 11)$. The configuration dies before any crossing is examined.

Lemma 5.4 (prime counting). *If b is prime, each residue r_i covers exactly the $2m+1$ multipliers $r_i^{-1}\{-m, \dots, m\}$; hence $b-1 \leq n(2m+1)$, and equality forces an exact tiling of $\mathbb{Z}_b^\times$ by the reflected inverse pairs $\pm r_i^{-1}$.*

5.2 The residual case load, completely enumerated

At $n = 4$ with $m = 1, \dots, 8$, all bridge values of b , and all configurations: exactly **116** configurations survive the orbit lemma. All 116 are killed by off-grid pair-sum crossings of their canonical lifts; none is realized. The $m = 1$ layer, in full, with killing witnesses:

$$\begin{array}{llll} b = 6 : \{1, 2, 3, 5\} & t = 1/4 & (u, w) = (1, 3), & \text{value } 1/4; & \{1, 3, 4, 5\} & t = 4/9 \\ b = 7 : \{1, 2, 3, 6\} & t = 1/4 & (u, w) = (1, 3), & \text{value } 1/4; & \{1, 2, 4, 6\} & t = 3/8 \\ b = 7 : \{1, 3, 5, 6\} & t = 5/11 & (u, w) = (5, 6), & \text{value } 3/11; & \{1, 4, 5, 6\} & t = 4/9 \\ b = 8 : \{1, 3, 4, 7\} & t = 1/5 & (u, w) = (1, 4), & \text{value } 1/5; & \{1, 4, 5, 7\} & t = 1/3 \end{array}$$

(every value exceeds the target $m/b = 1/6, \dots, 1/8$ by at least a factor $3/2$). The case count is completed by two structural laws.

The dilation law. The 20 survivors at $m = 2$ are all within $+1$ (per gap) of $2\times$ -dilations of the eight $m = 1$ patterns; the 10 survivors at $m = 3$ are all within $+1$ of $3\times$ -dilations; 8 of the 20 at $m = 2$ are exact dilations. Exact dilations carry identical witnesses (a dilated set is a scaled set, the gap is scale-invariant, and the killing time divides). The residual case load is therefore *parametric* in m : eight base patterns plus bounded perturbations — the “consecutive plus one modification” structure guessed in the previous turn’s open problem O1 is confirmed and is the whole story at $k = 3$.

The top-pair rescue law. For a bridge configuration surviving the orbit lemma, let $c_1 < \dots < c_n$ be any unit lift. Claim: the pair-sum crossing family of the two *largest* speeds (the residues nearest the down-pole $b - m$) contains a time at which the envelope exceeds m/b . Measured coverage over all orbit-surviving (configuration, lift) pairs:

n	4	5	6	7	8	9	10
(config, lift) pairs	1,798	3,006	8,208	15,212	5,280	6,966	7,194
top-pair coverage	92.8%	94.9%	97.1%	97.0%	97.7%	98.3%	98.6%

Table 3: Top-pair rescue coverage, rising with n . On the canonical lift the top pair kills 116/116 residual cases at $n = 4$ and 320/323 at $n = 5$; the three exceptions fall to bottom and mixed pairs. Whenever the top pair fails, at least two (typically three at $n = 4$; five to thirty-five at $n = 10$) other winning pairs exist.

5.3 Assembly status and the honest comparison

The residue proof architecture at $k = 3$ is:

$$\text{deficit law at } n = 4 = \underbrace{\text{Lemma 5.1}}_{2 \text{ lines}} + \underbrace{\text{Lemma 5.2}}_{3 \text{ lines}} + \text{top-pair rescue},$$

where the last term is the top-pair rescue for eight base patterns and their bounded perturbations. The first two lemmas are proved and n -uniform. The rescue is the single remaining gap: one uniform statement, verified for $m \leq 8$ and all unit lifts at $n = 4$, and with the measured coverage of Table 3 at every size.

Compared with the classical proofs at four speeds (and the Barajas–Serra machinery up to seven), which are covering/potential arguments with n -specific case analysis whose pigeonhole content is the union bound $b \leq n(2m+1)+1$: the residue chain does not re-derive them. Its two proved lemmas kill 99.8% of the $n = 4$ bridge with no case analysis at all, and the residual is a parametric family with a uniform witness rule that has no classical counterpart. But the work is not avoided — it is relocated into the single rescue statement, which is precisely the factor-2 obstruction in disguise. The verdict on D1 is therefore *between* “translation” and “content”: the reformulation yields a genuinely different, n -uniform proof *architecture* with measured covering power, and it localizes the difficulty to a thin, highly structured set — but completing the proof requires proving the rescue law.

6 Verdict and recommendation

- **D1 (residue proof at $k = 3$): not a pure translation, not a proof.** The orbit lemma is strictly stronger than classical covering (it kills what the union bound cannot reach); the divisor lemma and the top-pair rescue law are new; the residual is parametric. The factor-2 wall is *localized* — to the shallow bridge, $|d| \leq 3$, near-regular residue patterns — not removed.
- **D2 (counterexample hunt at $k = 8, 9$): clean hold.** 627,266 exhaustive sets, 155,382 adversarial evaluations, 143,892 bridge configurations: zero violations, zero exceptions, zero machinery discrepancies. The floor at $n \geq 8$ is regular-isolated.
- **Clothing versus body:** the residue clothing closes $\sim 94\%$ of the body at *every* size $n = 3, \dots, 10$; the remaining $\sim 6\%$ — the shallow bridge — is where LRC actually lives in this language.

The advisor’s decision rule (both tests neutral $\Rightarrow$ pivot) is not quite triggered: the tests returned mildly positive but unproved. Two moves are now well-posed.

R1 (one bounded residue-track mile). Prove the top-pair rescue lemma: for a bridge configuration with residues $c_1 < \dots < c_n < b$ (any unit lift), the family $\{j/(c_{n-1} + c_n)\}$ contains t with $\min_i \|c_i t\| > m/b$. Success completes the $k = 3$ proof and gives the architecture for $k = 4, 5$.

R2 (the pivot, sharpened). The orbit lemma is itself a covering statement in the group language: *the dilates jR of the residue set always meet the arc $[-m, m]$* . This is exactly the covering formulation of the kernel dictionary (bridge prospectus, Step 1). The group-language track and the residue track therefore meet at Lemma 5.1: additive-combinatorial bounds on multiplicative covering of arcs in $\mathbb{Z}_b$ would simultaneously strengthen the orbit lemma and import the dictionary’s first inequality. The pivot is a continuation by other means, aimed at the shallow bridge, with Table 2 as the target map.

7 Reproduction

All scripts live in `scripts/` (mirrored to `computational-check/ diag2/`) with outputs in `scripts/out/ diag2_*.json`: `lrc_diag_gap.py` (engine + validation), `lrc_diag_scan.py` (exhaustive scans), `lrc_diag_climb.py` (adversarial hill-climb), `lrc_diag_critical_hunt.py` (critical-set hunt), `lrc_diag_bridge.py` (bridge attribution), `lrc_diag_casebook.py` (complete residual case book at $n = 4, 5$), `lrc_diag_rescue.py` (top-pair rescue coverage). Run-times: the $n = 10$ scan is 52 s; everything else runs in seconds to a minute on two cores; exact rational arithmetic throughout.